

1) **INIETTIVA**:

$$\forall a, a' \in A \quad f(a) = f(a') \Rightarrow a = a'$$

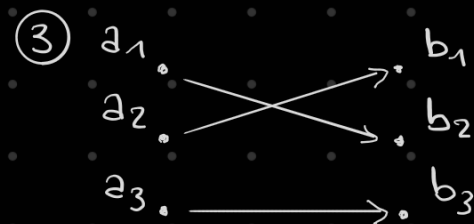
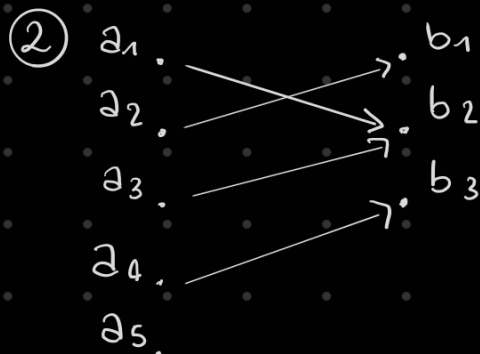
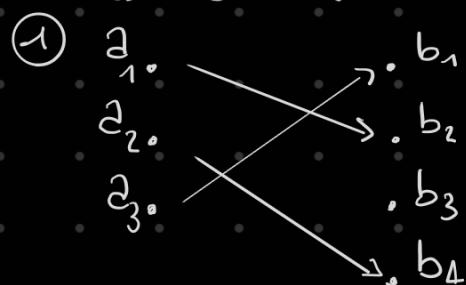
2) **SURIETTIVA**:

$$\text{Im } f = B \quad \text{ovvero}$$

$$\forall b \in B \quad \exists a \in A \mid f(a) = b$$

3) **BIETTIVA** (o BIUNIVOCA)

iniettiva e suriettiva insieme



Esercitazione 1

1) S_3 ha $3!$ elt:

$$e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

↘ identità ↗

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

↑↑ cicliche

se un numero va in se stesso non c'è bisogno di rappresentarlo: $(1, 2)$

tabella moltiplicativa:

$\cdot$	e	$(1, 2)$	$(2, 3)$	$(1, 3)$	$(1, 2, 3)$	$(1, 3, 2)$
e	e	$(1, 2)$	$(2, 3)$	$(1, 3)$	$(1, 2, 3)$	$(1, 3, 2)$
$(1, 2)$	$(1, 2)$	e	$(1, 2, 3)$	$(1, 2, 3)$	$(1, 2)$	$(1, 3, 2)$
$(2, 3)$	$(2, 3)$	$(1, 2, 3)$	e	$(1, 2, 3)$	$(2, 3)$	$(1, 2)$
$(1, 3)$	$(1, 3)$	$(1, 2, 3)$	$(1, 2, 3)$	e	$(1, 3)$	$(1, 3, 2)$
$(1, 2, 3)$	$(1, 2, 3)$	$(1, 3, 2)$	$(1, 3, 2)$	$(2, 3)$	e	e
$(1, 3, 2)$	$(1, 3, 2)$	$(2, 3)$	$(1, 3, 2)$	$(1, 2)$	$(1, 2, 3)$	$(1, 2, 3)$

dove zero essere giusto

parto sempre dallo stesso lato.

$$\begin{array}{ccc} 1 & 2 & 1 \\ 2 & \rightarrow 1 & \rightarrow 2 \Rightarrow e \\ 3 & 3 & 3 \end{array}$$

S_3 è generato da elementi σ, τ tali che:

$$\sigma^2 = 1 \quad e \quad \tau^3 = 1 \quad e$$

$$\sigma\tau = \tau^2\sigma = \sigma\tau\tau = \sigma\tau^2\tau = \sigma\tau^2\tau^2 = \sigma\tau^4 = \sigma\tau\tau^3 = \sigma\tau$$

2) $\sigma \circ \tau =$

1	2	3	4	5	6
1	3	2	4	6	5

3) $x y z = 1$, $y z x = 1$? $y x z = 1$?

$$\forall x, y, z \in G \text{ gruppo}$$

$$\begin{aligned}
 (xy)z = 1 &\Rightarrow (xy)zz^{-1} = 1z^{-1} \Rightarrow xy = z^{-1} \\
 &\Rightarrow xy y^{-1} = y^{-1}z^{-1} \Rightarrow x = y^{-1}z^{-1} \\
 &\Rightarrow xy z = y z y^{-1} z^{-1} \Rightarrow xy z = 1
 \end{aligned}$$

procedimento valido anche per l'altro caso.

4) $(G, \cdot), (G^o, \circ) \mid a \circ b = b \cdot a$. G^o gruppo?

$$\textcircled{1} a \circ (b \circ c) = (a \circ b) \circ c$$

$$\Rightarrow a \circ cb = ba \circ c \Rightarrow cba = cba$$

$$\textcircled{2} e \circ a = ae = a \wedge a \circ e = ea = a$$

$$\textcircled{3} a^{-1} \circ a = a \circ a^{-1} = e \quad \square$$

5) $a, b \in G$ gruppo. a ha ordine 5. $a^3 b = b a^3$. $ab = ba$?
 $\hookrightarrow a^5 = 1$

$$ab = 1ab \Rightarrow ab = a^5 ab \Rightarrow ab = a^6 b \Rightarrow ab = ba^6$$

$$\Rightarrow ab = ba^5 a \Rightarrow ab = ba \quad \square$$

6) a) si c) no e) no
 b) si d) si

$\rightarrow GL_n(\mathbb{K})$ è gruppo lineare delle $M_{n \times n}$ nell'insieme $M_{\mathbb{K}}$ invertibili.

l'inverso di $A \cdot B$ è $B^{-1} A^{-1}$

l'inversa di una $M_{2 \times 2}$ la trovo così

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ è invertibile se $ad - bc \neq 0$.

$$\Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

esercizio:

dimostrare per induzione che

$$\sum_{i=0}^n 4i+1 = (n+1) \cdot (2n+1)$$

$$\text{B.I.} \rightarrow n=0 \Rightarrow 1=1 \quad \checkmark$$

$$\text{P.I.} \rightarrow n=n+1$$

$$\sum_{i=0}^{n+1} 4i+1 = \sum_{i=0}^n 4i+1 + 4(n+1)+1 =$$

$$(n+1)(2n+1) + 4(n+1)+1 =$$

$$2n^2 + n + 2n + 1 + 4n + 4 + 1$$

$$2n^2 + 7n + 6$$

$$(n+2)(2n+2+1) = 2n^2 + 2n + n + 4n + 4 + 2 =$$

$$2n^2 + 7n + 6 \quad \square$$

es. siano $a, b, c \in \mathbb{Z}$ e $c|a$, $c|b$. verificare che:

$$\forall \alpha, \beta \in \mathbb{Z} \quad c | (\alpha a + \beta b)$$

ESERCITAZIONE 2

10) $(3522, 321)$ già fatto a lezione ✓

$$(413, 173)$$

$$413 = 173 \cdot 2 + 67$$

$$173 = 67 \cdot 2 + 39$$

$$67 = 39 \cdot 1 + 28$$

$$39 = 28 \cdot 1 + 11$$

$$28 = 11 \cdot 2 + 6$$

$$11 = 6 \cdot 1 + 5$$

$$6 = 5 \cdot 1 + 1$$

$$5 = 1 \cdot 5 + 0 \quad \Rightarrow \text{MCD} = 1$$

bezout:

$$1 = 6 - 5 \cdot 1$$

$$= 6 - (11 - 6 \cdot 1) = 6 \cdot 2 - 11$$

$$= (28 - 11 \cdot 2) \cdot 2 - 11 = 28 \cdot 2 - 11 \cdot 5$$

$$= 28 \cdot 2 - (39 - 28 \cdot 1) \cdot 5 = 28 \cdot 7 - 39 \cdot 5$$

$$= (67 - 39 \cdot 1) \cdot 7 - 39 \cdot 5 = 67 \cdot 7 - 39 \cdot 12$$

$$= 67 \cdot 7 - (173 - 67 \cdot 2) \cdot 12 = 67 \cdot 31 - 173 \cdot 12$$

$$= (413 - 173 \cdot 2) \cdot 31 - 173 \cdot 12 = 413 \cdot 31 - 173 \cdot 74$$

$$= 12803 - 12801 = 1$$

$$\Rightarrow \alpha = 31, \beta = -74$$

9) $b := x^3 - 6x^2 + x + 4$

$$a := x^5 - 6x + 1$$

$$\delta b(x) = 3 \quad \text{e} \quad \delta a(x) = 5 \quad \Rightarrow \quad \delta q(x) = 2 \quad \text{e} \quad \delta r(x) \leq 2$$

$$(x^5 - 6x + 1) = (x^3 - 6x^2 + x + 4)(ax^2 + bx + c) + (dx^2 + ex + f)$$

$$\Rightarrow \underline{ax^5} + \underline{bx^4} + \underline{cx^3} - \underline{6ax^4} - \underline{6bx^3} + \underline{6cx^2} + \underline{ax^3} + \underline{bx^2} + \underline{cx} + \underline{4ax^2} + \underline{4bx} + \underline{4c}$$
$$+ \underline{dx^2} + \underline{ex} + \underline{f}$$

$$\Rightarrow ax^5 + x^4(b - 6a) + x^3(c - 6b + a) + x^2(6c + b + 4a + d) + x(c + 4b + e) + (4c + f)$$

$$\begin{cases} a = 1 \\ b - 6a = 0 \end{cases}$$

$$\begin{cases} a = 1 \\ b = 6 \end{cases}$$

$$\Rightarrow \begin{cases} c - 6b + a = 0 \\ 6c + b + 4a + d = 0 \\ c + 4b + e = -6 \\ 4c + f = 1 \end{cases} \begin{cases} c = 35 \\ d = -220 \\ e = -65 \\ f = -139 \end{cases}$$

$$\Rightarrow q(x) = x^2 + 6x + 35 \quad e \quad r(x) = -220x^2 - 65x - 139$$

2) dimostra per induzione che

$$\sum_{i=1}^n 2i - 1 = n^2$$

$$\text{caso base: } n=1 \Rightarrow 2-1=1=1^2 \quad \checkmark$$

passo induttivo: $n+1$

$$\sum_{i=1}^{n+1} 2i - 1 = (n+1)^2$$

$$\Rightarrow \sum_{i=1}^n 2i - 1 + 2(n+1) - 1 = (n+1)^2$$

$$n^2 + 2n + 2 - 1 = n^2 + 2n + 1 \Rightarrow n^2 + 2n + 1 = n^2 + 2n + 1 \quad \checkmark$$

3) dimostrare che $\forall n \geq 1 \quad 4 \mid 5^n - 1$

$$\text{caso base: } n=1 \Rightarrow 5-1=4 \text{ divisibile per } 4.$$

passo induttivo: $n+1$

$$\Rightarrow 5^{n+1} - 1 \quad \text{servono i mod}$$

4) dimostrare per induzione che

$$\sum_{i=1}^n F_i^2 = F_n \cdot F_{n+1}$$

$$\text{sapendo } F_1 = F_2 = 1 \quad e \quad F_n = F_{n-1} + F_{n-2}$$

$$\text{caso base: } n=1 \Rightarrow F_1^2 = F_1 \cdot F_2 = 1 \cdot 1 = 1$$

passo induttivo: $n+1$

$$\Rightarrow \sum_{i=1}^{n+1} F_i^2 = F_{n+1} \cdot F_{n+2}$$

$$\sum_{i=1}^n F_i^2 + F_{n+1}^2 = F_{n+1} \cdot F_{n+2}$$

$$\Rightarrow F_n \cdot F_{n+1} + F_{n+1}^2 = F_{n+1} \cdot F_{n+2}$$

$$F_{n+1} (F_n + F_{n+1}) = F_{n+1} \cdot F_{n+2}$$

$$F_n + F_{n+1} = F_{n+2}$$

$$F_{n+2} = F_{n+2} \quad \square$$

$$7) \begin{cases} 2x \equiv 3 \pmod{15} \\ 3x \equiv 5 \pmod{8} \\ 4x \equiv 2 \pmod{7} \end{cases} \begin{cases} x \equiv 9 \pmod{15} \\ x \equiv 7 \pmod{8} \\ x \equiv 4 \pmod{7} \end{cases}$$

$$2 \cdot 8 = 16 \equiv 1 \pmod{15} \Rightarrow 8 \cdot 3 = 24 \equiv 9 \pmod{15}$$

$$3 \cdot 11 = 33 \equiv 1 \pmod{8} \Rightarrow 11 \cdot 5 = 55 \equiv 7 \pmod{8}$$

$$4 \cdot 9 = 36 \equiv 1 \pmod{7} \Rightarrow 9 \cdot 2 = 18 \equiv 4 \pmod{7}$$

ESERCITAZIONE 3

1) $(14743, 8915)$

$$14743 = 8915 \cdot 1 + 5828$$

$$8915 = 5828 \cdot 1 + 3087$$

$$5828 = 3087 \cdot 1 + 2741$$

$$3087 = 2741 \cdot 1 + 346$$

$$2741 = 346 \cdot 7 + 319$$

$$346 = 319 \cdot 1 + 27$$

$$319 = 27 \cdot 11 + 22$$

$$27 = 22 \cdot 1 + 5$$

$$22 = 5 \cdot 4 + 2$$

$$5 = 2 \cdot 2 + 1$$

$$2 = 1 \cdot 2 + 0$$

$$\Rightarrow \text{MCD} = 1$$

bezout:

$$1 = 5 - 2 \cdot 2$$

$$= 5 - (22 - 5 \cdot 4) \cdot 2 = 5 \cdot 9 - 22 \cdot 2$$

$$= -22 \cdot 2 + (27 - 22 \cdot 1) \cdot 9 = 27 \cdot 9 - 22 \cdot 11$$

$$= 27 \cdot 9 - (319 - 27 \cdot 11) \cdot 11 = 27 \cdot 130 - 319 \cdot 11$$

$$= (346 - 319 \cdot 1) \cdot 130 - 319 \cdot 11 = 346 \cdot 130 - 319 \cdot 141$$

$$= 346 \cdot 130 - (2741 - 346 \cdot 7) \cdot 141 = 346 \cdot 1117 - 2741 \cdot 141$$

$$= (3087 - 2741 \cdot 1) \cdot 1117 - 2741 \cdot 141 = 3087 \cdot 1117 - 1268 \cdot 2741$$

$$= 3087 \cdot 1117 - (5828 - 3087 \cdot 1) \cdot 1258 = 2375 \cdot 3087 - 5828 \cdot 1258$$

$$= (8915 - 5828 \cdot 1) \cdot 2375 - 5828 \cdot 1258 = 8915 \cdot 2375 - 3633 \cdot 5828$$

$$= 8915 \cdot 2375 - (14743 - 8915 \cdot 1) \cdot 3633 = 8915 \cdot 6008 - 14743 \cdot 3633$$

$$\alpha = -3633 \quad \beta = 6008$$

$(10000, 642)$

$$10000 = 642 \cdot 15 + 370$$

$$642 = 370 \cdot 1 + 272$$

$$370 = 272 \cdot 1 + 98$$

$$272 = 98 \cdot 2 + 76$$

$$98 = 76 \cdot 1 + 22$$

$$76 = 22 \cdot 3 + 10$$

$$22 = 10 \cdot 2 + 2$$

$$10 = 2 \cdot 5 + 0 \quad \Rightarrow \text{MCD} = 2$$

bezout:

$$2 = 22 - 10 \cdot 2$$

$$= 22 - (76 - 22 \cdot 3) \cdot 2 = 22 \cdot 7 - 76 \cdot 2$$

$$= (98 - 76 \cdot 1) \cdot 7 - 76 \cdot 2 = 98 \cdot 7 - 76 \cdot 9$$

$$= 98 \cdot 7 - (272 - 98 \cdot 2) \cdot 9 = 98 \cdot 25 - 272 \cdot 9$$

$$= (370 - 272 \cdot 1) \cdot 25 - 272 \cdot 9 = 370 \cdot 25 - 272 \cdot 34$$

$$= 370 \cdot 25 - (642 - 370 \cdot 1) \cdot 34 = 370 \cdot 59 - 642 \cdot 34$$

$$= (10000 - 642 \cdot 15) \cdot 59 - 642 \cdot 34 = 10000 \cdot 59 - 642 \cdot 919$$

$$\alpha = 59 \quad \beta = -919$$

$$(5785, 546)$$

$$5785 = 546 \cdot 10 + 325$$

$$546 = 325 \cdot 1 + 221$$

$$325 = 221 \cdot 1 + 104$$

$$221 = 104 \cdot 2 + 13$$

$$104 = 13 \cdot 8 + 0 \quad \Rightarrow \text{MCD} = 13$$

bezout:

$$13 = 221 - 104 \cdot 2$$

$$= 221 - (325 - 221 \cdot 1) \cdot 2 = 221 \cdot 3 - 325 \cdot 2$$

$$= (546 - 325 \cdot 1) \cdot 3 - 325 \cdot 2 = 546 \cdot 3 - 325 \cdot 5$$

$$= 546 \cdot 3 - (5785 - 546 \cdot 10) \cdot 5 = 546 \cdot 53 - 5785 \cdot 5$$

$$\alpha = -5 \quad \beta = 53$$